EC 571: Digital VLSI Circuit Design

Overview of Semiconductor Memory

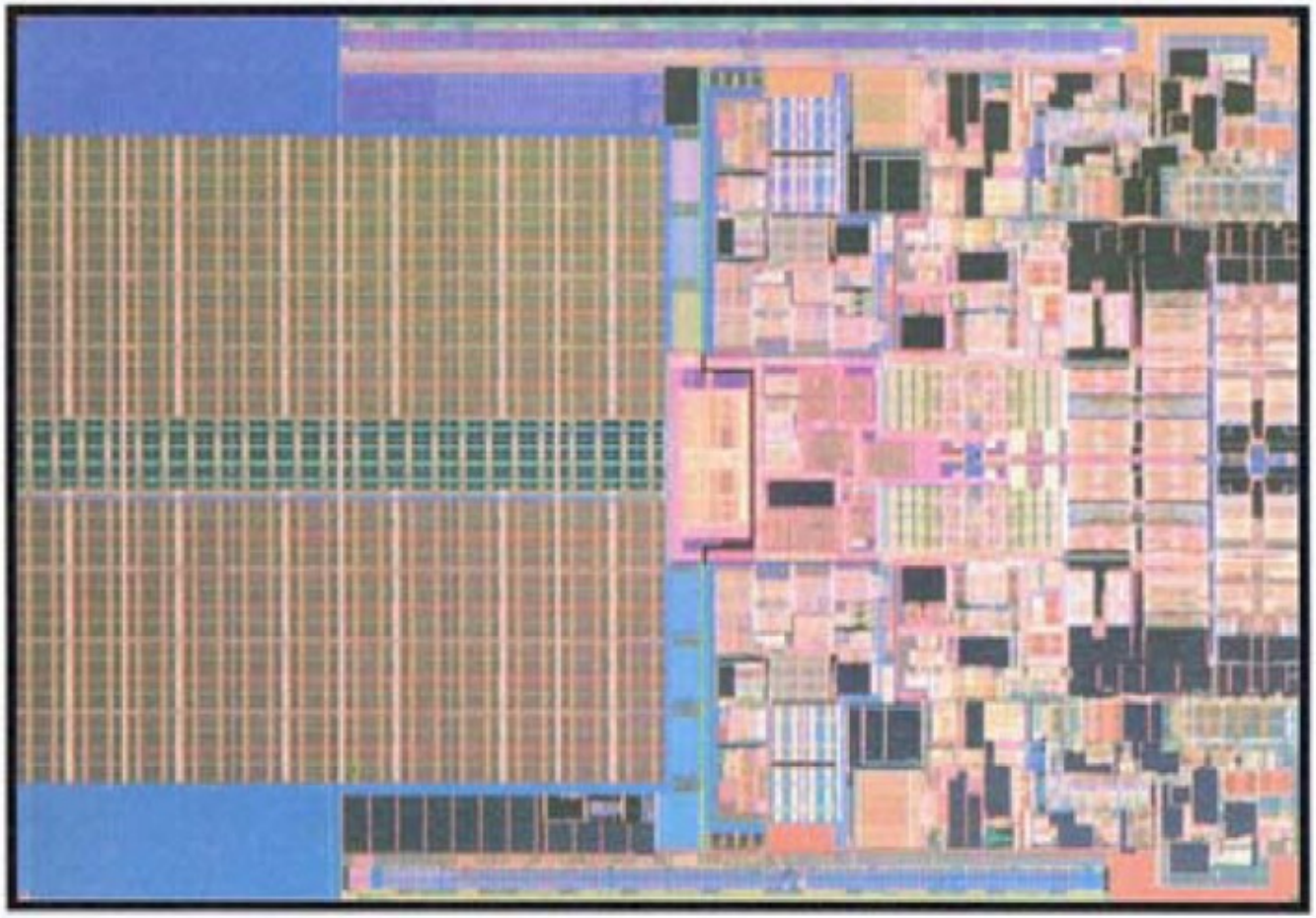
Ajay Joshi

Acknowledgement - Some slides adapted from:
- J. Rabaey et al. Digital Integrated Circuits 2nd Ed.
- UC Berkeley EE141, UC Berkeley EE241, UC Berkeley EE251
- Stanford EE271

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Memory design

Intel 45nm Core 2



Semiconductor Memory Classification

Volatile Read-Write Memory		Non-Volatile Read-Write Memory	Non-Volatile Read-Only Memory
Random Access	**Non-Random Access**	EPROM E^2PROM FLASH	Programmable (PROM)
SRAM DRAM	FIFO LIFO Shift Register CAM		

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Random Access Memories (RAM)

- Static RAM
  - Data stored as long as supply is applied
  - Larger (6 transistors/cell)
  - Fast
  - Differential (usually)
- Dynamic RAM
  - Periodic refresh required
  - Smaller (1-3 transistors/cell)
  - Slower
  - Single-ended

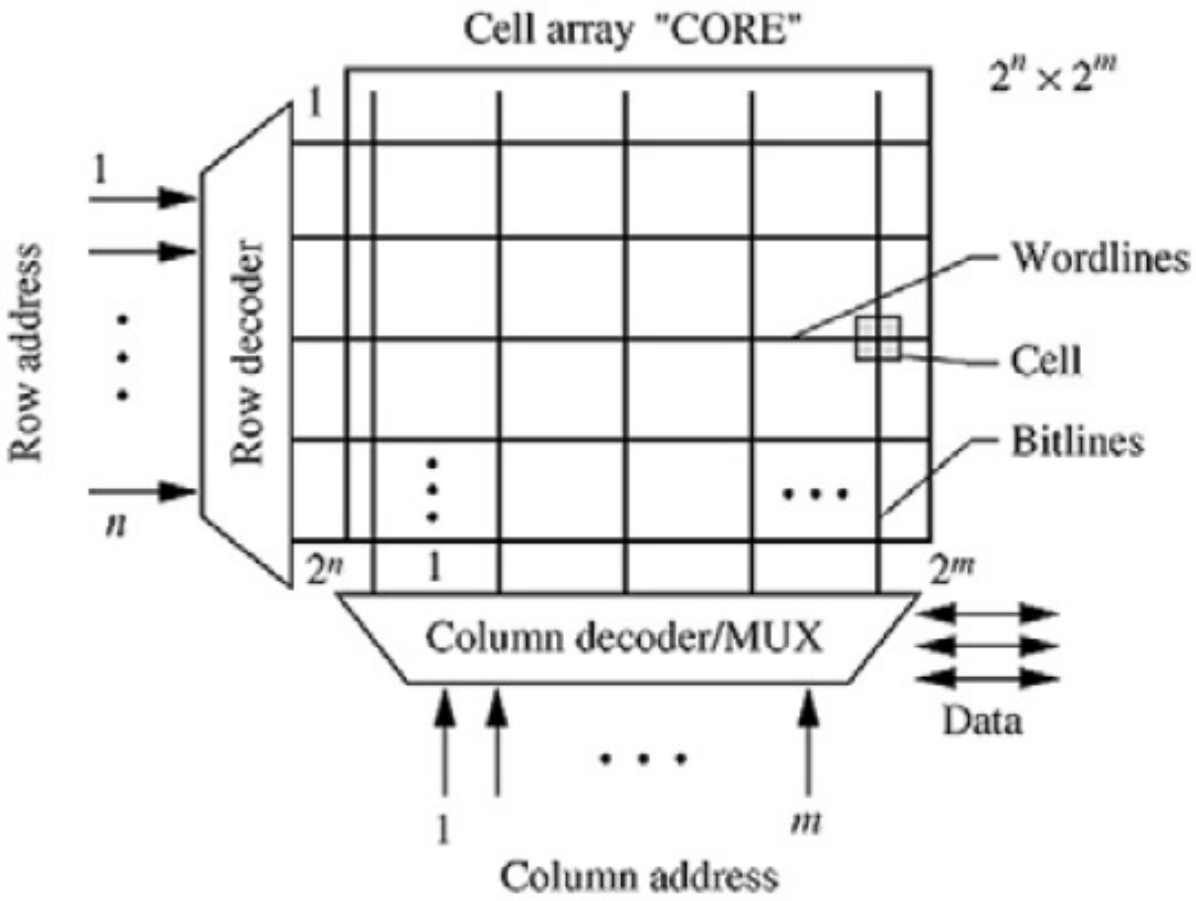
Basic memory array

CORE:
- keep square within a 2:1 ratio
- rows are **word lines**
- columns are **bit lines**
- data in and out on columns

DECODERS:
- needed to reduce total number of pins; N+M address lines for 2^{N+M} bits of storage

Ex: if N+M=20 → 2^{20}= 1Mb

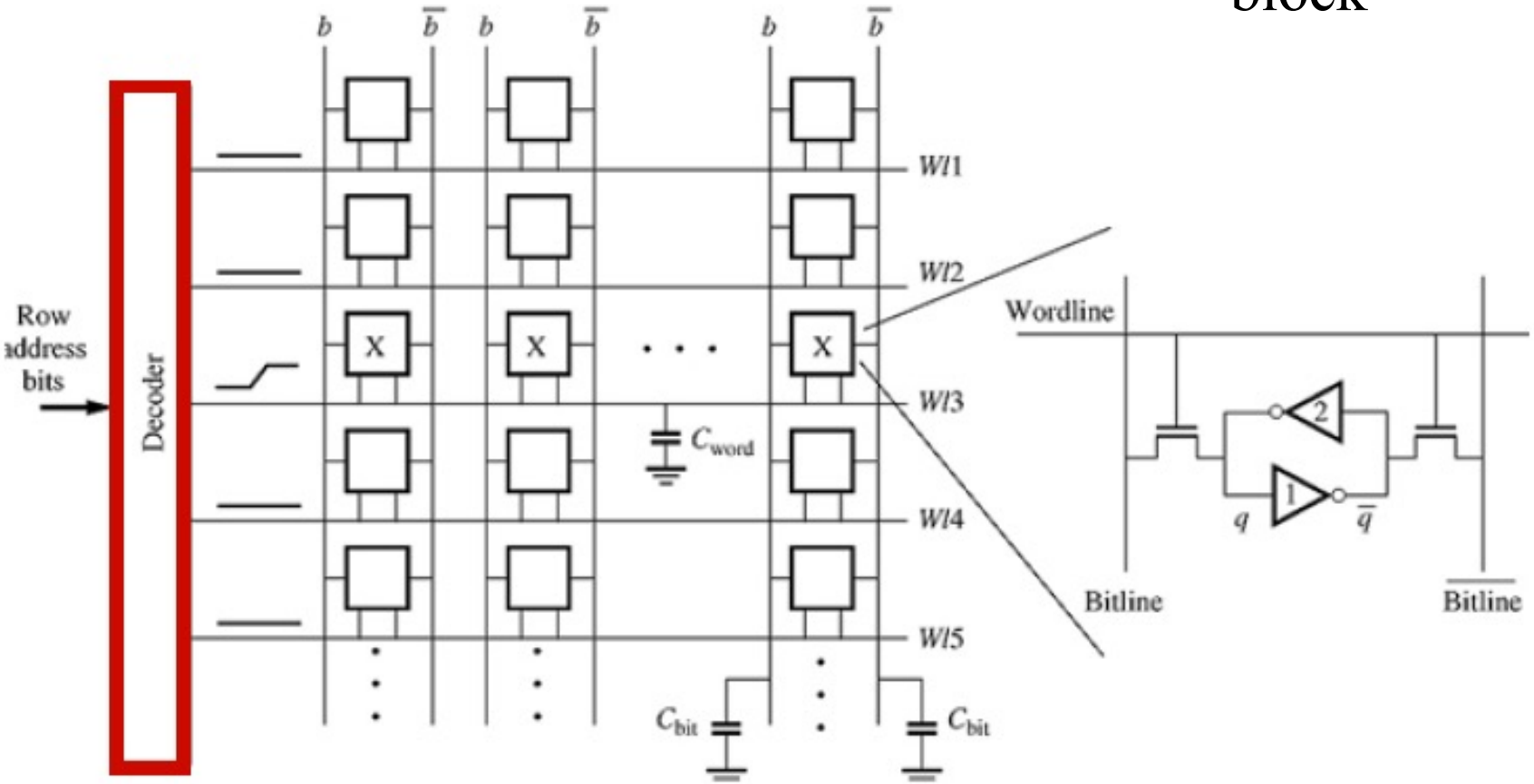
MULTIPLEXING:
- used to select one or more columns for input or output of data



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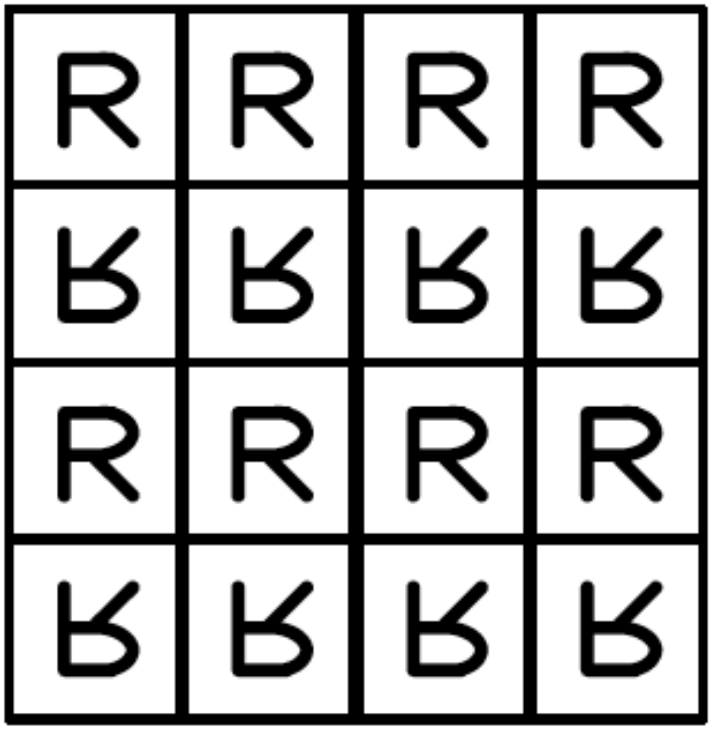
SRAM array

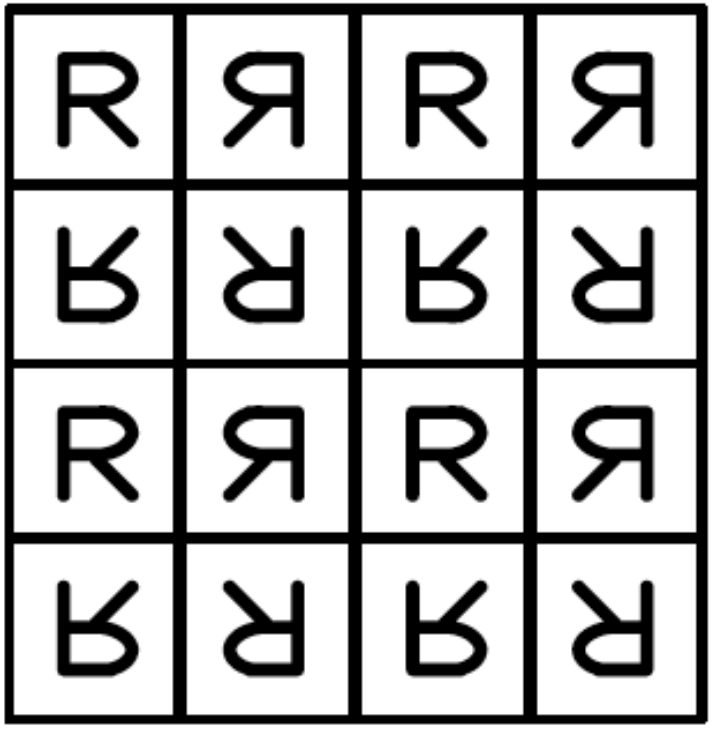
256 x 256 memory block



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SRAM array layout

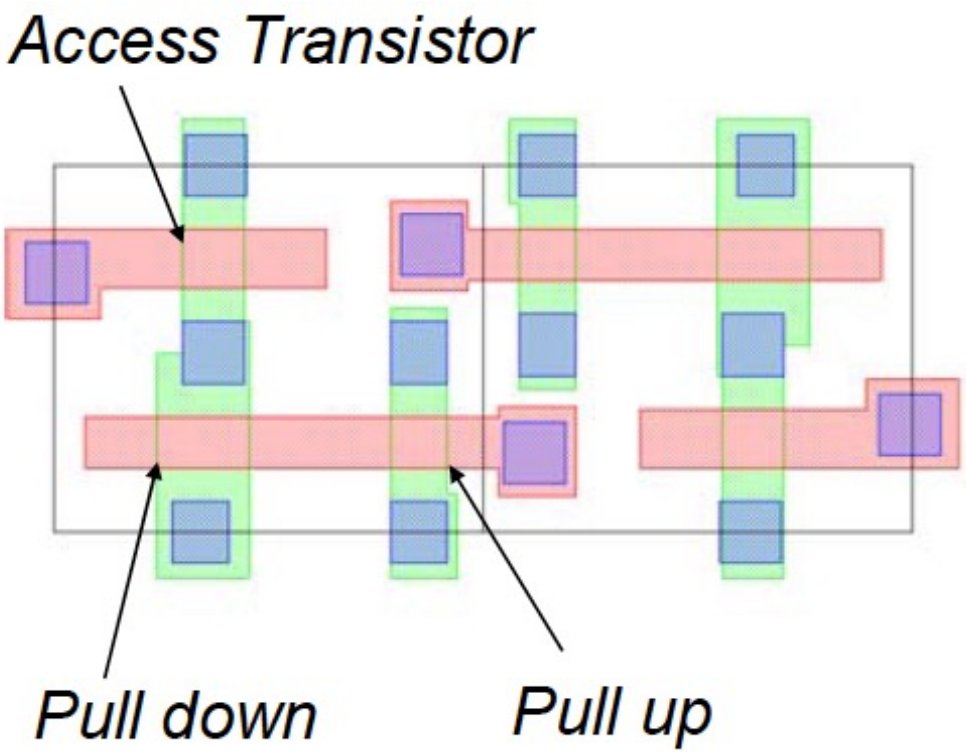


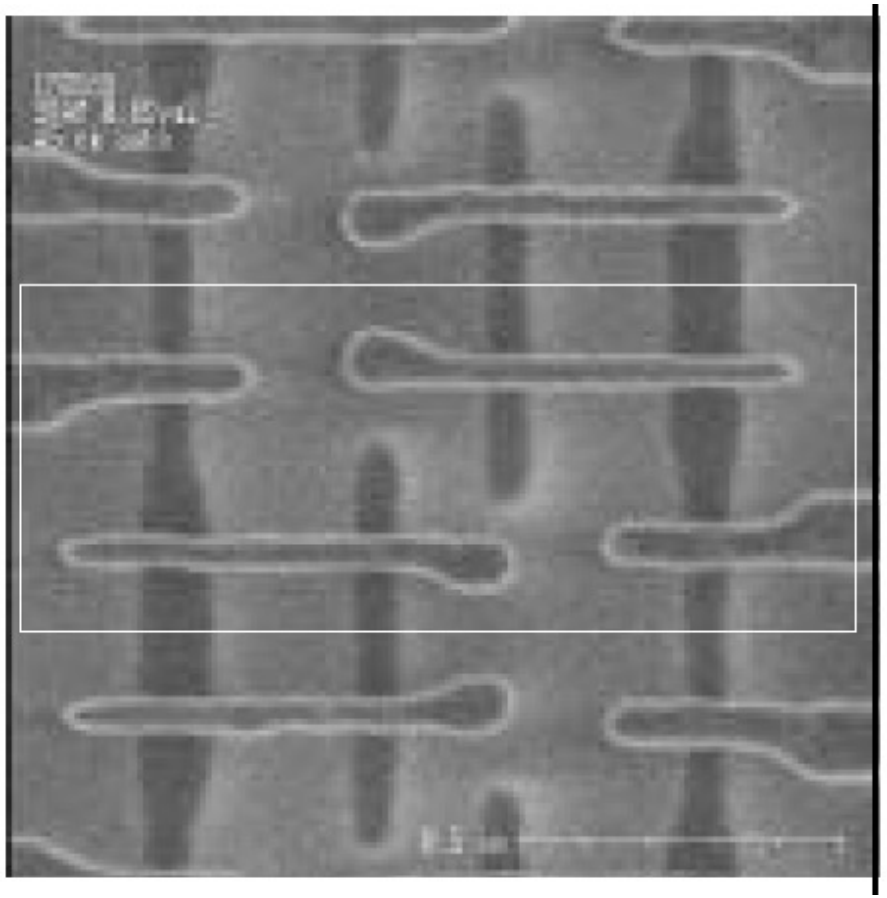


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65 nm SRAM cell

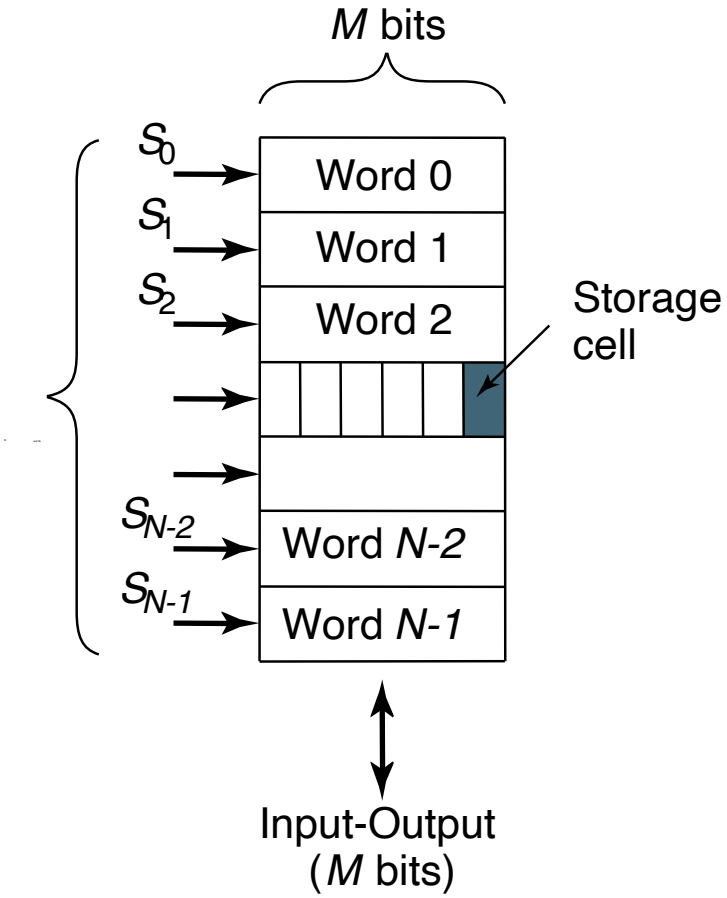
- ST/Philips/Motorola

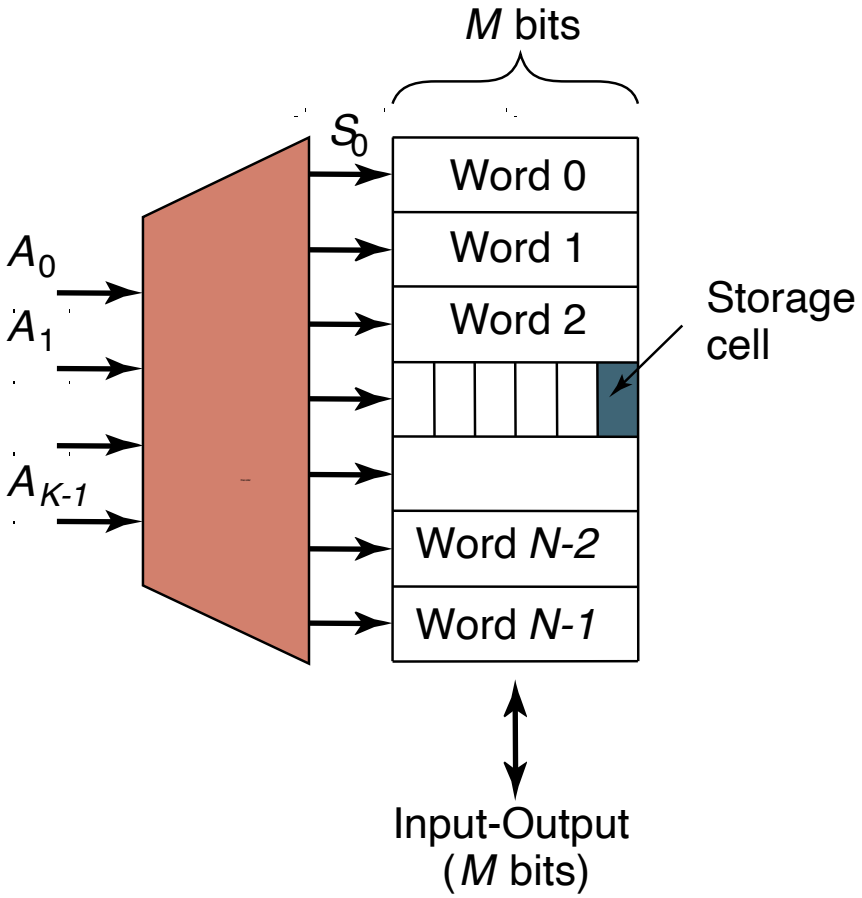




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Memory Architecture: Decoder design





Intuitive architecture for N x M memory
Too many select signals:
N words == N select signals

Decoder reduces the number of select signals

$$K = log_2 N$$

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